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# Hybrid green nonaqueous media: tetraethylene glycol modifies the properties of a (choline chloride + urea) deep eutectic solvent†

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Deep eutectic solvents (DESs) have emerged as easy-to-prepare inexpensive environmentally benign media with the potential for applications in various areas of chemistry. A judiciously selected cosolvent can modify the properties of a DES in a favorable manner. Tetraethylene glycol (TEG)-modified DES composed of choline chloride and urea in a 1 : 2 molar ratio, named Reline, is investigated over the complete composition regime within 298–358 K for its properties. The empirical solvent polarity parameter, $E_T^N$, obtained using the solvatochromic absorbance probe response of a betaine dye, along with Kamlet–Taft parameters, dipolarity/polarizability ($\pi^*$), H-bond donating acidity ($\alpha$), and H-bond accepting basicity ($\beta$) suggest the presence of interactions, mainly H-bonded, between TEG and the components of Reline. The H-bond accepting basicity ($\beta$) of TEG-modified Reline is found to be unusually high. Responses from dipolarity as well as intramolecular charge-transfer fluorescence probes further support these outcomes, suggesting that a small amount of TEG can effectively alter the properties of Reline. Negative molar excess volumes and positive excess logarithmic viscosities estimated from density and dynamic viscosity measurements, respectively, at all compositions and temperatures for (Reline + TEG) mixtures indicate the presence of stronger inter-species H-bonding interactions between Reline–TEG as compared to the intra-species H-bonding interactions between Reline–Reline and between TEG–TEG. FTIR absorbance and Raman spectral measurements indicate unusually high H-bond accepting basicity ($\beta$) of the (Reline + TEG) mixture to be due to the decreased involvement of urea functionalities in H-bonding with choline chloride, with possible increase in H-bonding between the adequate functionalities of TEG with those of choline chloride.

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## 1. Introduction

Deep eutectic solvents (DESs) are rapidly gaining interest from both academic and industrial research communities worldwide.[1–7] The reason for the increased interest in DESs lies in the favorable features and properties associated with these novel, environmentally-benign, easy-to-prepare, and inexpensive solvents. Most of the common DESs, especially the ones constituted by combining quaternary ammonium salts and common hydrogen bond donors, have been documented to be nontoxic, a feature that renders them superior to their ionic liquid counterparts.[1–3] These DESs are also cost-effective as the starting materials, common quaternary ammonium salts (*e.g.*, choline chloride) and hydrogen bond donors (*e.g.*, urea, glycerol, *etc.*) that are combined to form this class of DESs, are fairly inexpensive, and the synthetic strategy to prepare these DESs is also simple, straight-forward, and less energy-intensive.[1–5] The interest in DESs from academia also arises from the fact that many of these sustainable liquids are prepared by simply mixing two or more solids due to the interesting modifications in the intermolecular interactions within the system as a result of the mixing.

The appropriate combination of the constituents of DESs turns them into solvent milieu of anomalous behavior that is remarkably different from the common molecular solvents or aqueous/organic electrolytes.[2–5] Apart from using different combinations of salt and hydrogen bond donor to prepare a DES with modified features, the physicochemical properties of DESs may also be effectively modified by simple addition of a cosolvent. In this context, in order to prepare nonaqueous solubilizing milieu composed of DES as a constituent, an environmentally-benign cosolvent, tetraethylene glycol (TEG), turns out to be a logical and interesting choice as it is constituted of termini hydroxyl as well as ethoxy functionalities that can offer both H-bond donating and H-bond accepting possibilities. Thus, TEG as a cosolvent has potential to impart favorable and unusual properties to DESs as solubilizing milieu. Further, the TEG is not only inexpensive, easily available, and

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easy to handle, it also shows appreciable miscibility with some of the common and popular DESs. TEG also shows several industrial applications especially as an intermediate and/or ingredient in polyester resins, as component of antifreezes/coolants, as plasticizers for nitrocellulose finishes, as humectants in tobacco and textiles, as lubricant for rubber, as heat transfer fluids, and in gas dehydration and treatment.[8,9] Due to its attractive properties and applications combined with H-bond donating/accepting capabilities, TEG clearly has potential to form a hybrid system with DES possessing attractive physicochemical properties.

In this work, we have observed that TEG can alter/modify properties of a typical and common DES, named Reline, prepared my mixing choline chloride and urea, both solids under ambient conditions, in 1 : 2 mole ratio. Several physical properties as well as solute solvation behavior of the colorless transparent liquid Reline under ambient conditions have been recently reported.[10–14] Apart from physical properties, solute solvation is of utmost importance as far as characterizing a complex solubilizing milieu is concerned. Information regarding the dependence of the solvation process on molecular architecture and functionalities of the solute is essential to establish any milieu as acceptable solubilizing medium in chemical sciences. Solvatochromic solutes (or probes), in this respect, may provide systematic information regarding the properties (*e.g.*, dipolarity, hydrogen bonding ability, *etc.*) of a complex solubilizing milieu.[15–18] Apart from the obvious solute–solvent interactions, solvent–solvent interactions due to alteration/modification in the medium may also get reflected in a solvatochromic probe response. Our investigation reported here reveals that addition of small amount of TEG modifies properties of Reline in a significant manner affording a new hybrid green nonaqueous media.

# 2. Experimental section

## 2.1. Materials

DES Reline was purchased in highest purity from Scionix Ltd. and stored in inert environment before its use. Alternatively, Reline was prepared by mixing choline chloride (≥99% from Sigma-Aldrich) and urea (≥99% from Sigma-Aldrich), respectively, in a mole ratio of 1 : 2 followed by stirring under heating (~353 K) until a homogeneous, colorless liquid has been formed. All spectroscopic measurements on DES purchased from Scionix Ltd. and those prepared by mixing choline chloride with urea were found to be statistically identical. TEG of 99% purity was purchased from Sigma-Aldrich. 2,6-Dichloro-4-(2,4,6-triphenyl-*N*-pyridino)phenolate (betaine dye 33) and 2,6-diphenyl-4-(2,4,6-triphenyl-*N*-pyridino)phenolate (betaine dye 30) were purchased in the highest available purity from Fluka (≥99%, HPLC). 4-Nitroaniline (NA) and *N*,*N*-diethyl-4-nitroaniline (DENA) were purchased in the highest purity from Spectrochem Co. Ltd. and Frinton Laboratories, respectively. Pyrene (Py), pyrene-1-carboxaldehyde (PyCHO), and coumarin-153 (C-153) were obtained in the highest purities from Sigma-Aldrich and were used as received. 6-*p*-Toluidine-2-naphthalenesulfonic acid (TNS) and 1-anilinonaphthalene-8-sulfonate (ANS) were obtained in the highest purities from Acros Organics. *N*,*N*-Dimethyl-6-propionyl-2-naphthylamine (PRODAN) was obtained from Biochemika in the highest purity and was used as received. Absolute ethanol was used to prepare probe stock solutions.

## 2.2. Methods

Stock solutions of all probes were prepared by dissolving in ethanol in pre-cleaned amber glass vials and stored at 4 ± 1 °C. The required amount of a probe was weighed using a Mettler-Toledo AB104-S balance with a precision of ±0.1 mg. An appropriate amount of the probe solution from the stock was transferred to a 1 $cm^2$ quartz cuvette. Ethanol was evaporated using a gentle stream of high purity nitrogen gas. (Reline + TEG) mixtures were prepared by mixing pre-calculated amount of Reline and TEG. The mole fraction of TEG ($x_{TEG}$) is estimated by taking the combination of 1 mol of choline chloride and 2 mol of urea as 1 mol of Reline. The mixture was heated at 363 K with gentle stirring for 30 minutes until it shows completely miscibility and forms one phase. Pre-calculated amount of the mixture cooled to room temperature is directly added to the cuvette and the solution is thoroughly mixed. The solubility of a probe within the mixture is checked using the linearity of the absorbance and/or the fluorescence intensity *versus* the concentration plot(s).

A Perkin-Elmer Lambda 35 double beam spectrophotometer with variable bandwidth and Peltier-temperature controller is used for acquisition of the UV-vis molecular absorbance data. Steady-state fluorescence spectra were acquired on a Jobin-Yvon Fluorolog-3 (model FL-3-11) modular spectrofluorometer equipped with a 450 W Xe arc lamp as the excitation source and single-grating monochromators as wavelength selection devices with photomultiplier tube as the detector. The temperature was controlled with a Thermo NESLAB RTE7 circulating chiller bath having a stability of ±0.01 °C. All absorbance and fluorescence data were acquired using 1 $cm^2$ quartz cuvettes. All fluorescence probes used were found to have adequate fluorescence quantum yields within the mixtures under investigation. Attenuated and reflectance-Fourier-transform infrared (ATR-FTIR) absorbance data were acquired from 4000 to 400 $cm^{-1}$ on an Agilent Technologies Cary 660 ATR double-beam spectrophotometer. The liquid samples were evenly spread on KBr pellets to record the FTIR absorbance spectra. Raman spectra were acquired with 532 nm excitation using a model no. X/01/220 XploRA PLUS Confocal Raman spectrometer. Densities ($\rho$) of the mixtures were measured using a Mettler Toledo, DE45 delta range density meter. The density measurement with the above mentioned density meter was based on electromagnetically-induced oscillations of a U-shaped glass tube. The standard deviations associated with the density measurement are ≤0.00005 g $cm^{-3}$. The measurements were performed at 15 K intervals in the temperature range of 298.15 K to 358.15 K. The dynamic viscosities ($\eta$) were measured with a Peltier-based (resolution of 0.01 K and accuracy < 0.05 K) automated Anton Paar microviscometer (model AMVn) having calibrated glass capillaries of different diameters (1.6, 1.8, 3.0, and

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4.0 mm). This instrument is based on the rolling-ball principle, where the steel ball rolls down the inside of inclined, sample-filled calibrated glass capillaries. The deviation in $\eta$ was $\leq$0.5%.

All measurements were performed at least in triplicate starting from the sample preparation and the results were averaged. All spectra were duly corrected by measuring the spectral responses from suitable blanks prior to data analysis and statistical treatment.

# 3. Results and discussion

### 3.1. Solvatochromic absorbance probes: response of Reichardt's dye and Kamlet–Taft parameters

A class of betaine dyes is well-known to exhibit remarkable negative solvatochromism as the decrease in the polarity of the cybotactic region results in a large bathochromic shift in the lowest energy intramolecular charge-transfer absorbance band of the dye (structures of the betaine dyes used in this investigation are presented in Scheme 1).[19–21] It is established that the Kamlet–Taft parameters dipolarity/polarizability ($\pi^*$) in combination with H-bond donating (HBD) acidity ($\alpha$) strongly influence this negative solvatochromism.[22–24] The $E_T(30)$ empirical scale of solvent polarity is defined as the molar transition energy of betaine dye 30 in kcal mol$^{-1}$ under ambient conditions according to the expression $E_T(30) = 28\,591.5/\lambda_{max}^{abs}$ (nm). As the betaine dye 30 does not exhibit appreciable solubility in DES Reline, another dye of this class betaine dye 33 (a halogenated derivative of betaine dye 30, Scheme 1) is used in the present study. The lowest energy absorbance transition of this dye [*i.e.*, $E_T(33)$] is calculated the same way $E_T(30)$ is calculated.[25,26] The $E_T(33)$ can be converted to normalized $E_T^N$ values using eqn (1) and (2).[14,21]

$$E_T(30) = 0.9953(\pm 0.0287)E_T(33) - 8.1132(\pm 1.6546) \quad (1)$$

$$R = 0.9926, \text{ standard error of estimate} = 0.8320, n = 20$$

$$E_T^N = \frac{[E_T(30)_{solvent} - E_T(30)_{TMS}]}{[E_T(30)_{water} - E_T(30)_{TMS}]} = \frac{[E_T(30)_{solvent} - 30.7]}{32.4} \quad (2)$$

here, TMS stands for tetramethylsilane, and $E_T(30)_{water} = 63.1$ kcal mol$^{-1}$ and $E_T(30)_{TMS} = 30.7$ kcal mol$^{-1}$ are experimentally observed values. $E_T^N$ is easier to conceive as it is dimensionless and varies between 0 for TMS (extreme nonpolar) and 1 for water (extreme polar). In order to de-convolve the roles of dipolarity/polarizability ($\pi^*$) and HBD acidity ($\alpha$), these empirical Kamlet–Taft parameters along with H-bond accepting (HBA) basicity ($\beta$) were also estimated.[22–24] The $\pi^*$ was estimated from the absorption maxima of DENA ($\bar{\nu}_{DENA}$ in kK, where kK = $10^3$ cm$^{-1}$), a non-HBD solute, using the following expression.[22–24]

$$\pi^* = 8.649 - 0.314\bar{\nu}_{DENA} \quad (3)$$

The $\alpha$ was determined from $E_T(30)$ and $\pi^*$ using:[22–24]

$$\alpha = \left[\frac{E_T(30) - 14.6(\pi^* - 0.23\delta) - 30.31}{16.5}\right] \quad (4)$$

and the $\beta$ was estimated from the enhanced solvatochromic shift of NA absorbance band maxima relative to its homomorph DENA, $-\Delta\bar{\nu}$(DENA − NA)/kK, according to:[22–24]

$$\beta = -0.357\bar{\nu}_{NA} - 1.176\pi^* + 11.12 \quad (5)$$

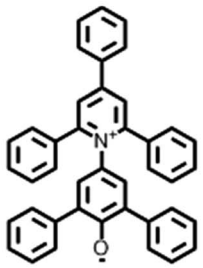
$E_T^N$ along with $\pi^*$, $\alpha$, and $\beta$ were estimated from the absorbance band maxima of betaine dye 33, DENA, and NA using eqn (1)–(5) for (Reline + TEG) mixtures at several different mixture compositions in the temperatures range 298–358 K at 15 K interval. While the numerical values of $E_T^N$, $\pi^*$, $\alpha$, and $\beta$ are reported in Table S1;† the representative absorbance spectra of betaine dye 33, DENA, and NA are presented in Fig. 1 and the plots of $E_T^N$, $\pi^*$, $\alpha$, and $\beta$ as a function of TEG mole fraction ($x_{TEG}$) are presented in Fig. 2. The parameters obtained for neat Reline and neat TEG are in good agreement with those reported earlier in literature.[13,27]

It is interesting to note that, irrespective of the temperature, $E_T^N$, $\pi^*$, and $\alpha$ in neat Reline are all higher than their corresponding values in neat TEG. A careful examination of the data reveals this difference to be more for $\pi^*$ and less for $\alpha$ with $E_T^N$, as expected, representing the net effect [see eqn (3) and (4)]. The fact that Reline is partly constituted of ions may be the reason for the relatively higher dipolarity/polarizability of this DES as compared to TEG. Although the presence of termini hydroxyl functionalities impart appreciable HBD acidity to TEG, urea and the hydroxyl group on choline cation both act as H-bond donors in Reline resulting in a slightly higher overall HBD acidity of Reline over TEG. The combination of termini hydroxyl along with the presence of the ethoxy functionalities of TEG, however, does result in significantly higher HBA basicity for neat TEG as compared to neat Reline irrespective of the temperature.

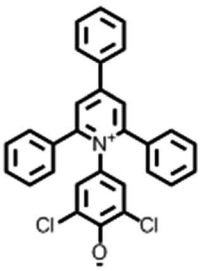
As expected, addition of TEG to Reline results in decrease in $E_T^N$, $\pi^*$, and $\alpha$, and increase in $\beta$ within the temperature range investigated. A careful examination of the plots in Fig. 2 reveals that as small amounts of TEG is added to Reline, $E_T^N$ decrease considerably and are found to be less than those predicted from ideal-additive behavior (shown using dashed/dotted lines) in the Reline-rich region. Whereas in TEG-rich region, experimental $E_T^N$ are slightly higher than the predicted ideal-additive value. Mixture of (Reline + TEG) around equimolar composition shows ideal additive behavior. The trend of decrease in $\pi^*$ and $\alpha$, respectively, suggest that the lower than expected $E_T^N$ in Reline-rich region is due to the significantly reduced HBD acidity ($\alpha$) as TEG is added to Reline. It is already established that $\alpha$ contributes ~68% towards $E_T^N$.[21,28,29] The higher than expected $E_T^N$ in TEG-rich region is also due to the unusually higher $\alpha$ in this region. The experimental dipolarity/polarizability ($\pi^*$) are more-or-less close to their ideal-additive values irrespective of the mixture composition and the temperature. It is interesting to note that while $\alpha$ are observed to be lower and higher than their expected values in Reline-rich and TEG-rich compositions, respectively, the HBA basicity ($\beta$) are found to be considerably higher than those predicted from the ideal-additive behavior irrespective of mixture composition and temperature. While TEG shows higher $\beta$ as compared to Reline,

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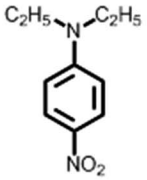
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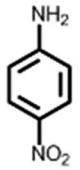
2,6-Diphenyl-4-(2,4,6-triphenyl-*N*-pyridino)phenolate (betaine dye 30)



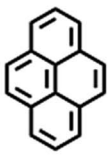
2,6-Dichloro-4-(2,4,6-triphenyl-*N*-pyridino)phenolate (betaine dye 33)



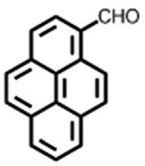
*N*,*N*-Diethyl-4-nitroaniline (DENA)



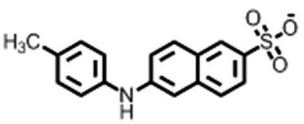
4-Nitroaniline (NA)



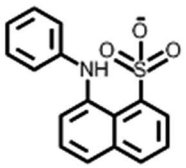
Pyrene (Py)



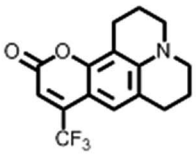
Pyrene-1-carboxaldehyde (PyCHO)



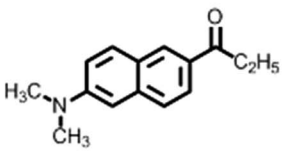
2-(*p*-Toluidino)naphthalene-6-sulfonate (TNS)



8-Anilino-1-naphthalene-sulfonate (ANS)



Coumarin-153 (C-153)



2-*N*,*N*-(Dimethylamino)-6-propionylnaphthalene (PRODAN)

Scheme 1 Structure of the absorbance and fluorescence probes used.

surprisingly, for the mixture composition $0.4 \leq x_{TEG} \leq 0.6$, HBA basicity ($\beta$) are found to be even higher than those observed in neat TEG. The unusually high HBA basicity associated to the (Reline + TEG) mixture is rare and a consequence of extensive changes in the H-bonding network when Reline and TEG are mixed. This is in complete contrast to the TEG mixture of several common ionic liquids constituted of imidazolium family HBD cations, where HBD acidity ($\alpha$) was, in general, found to be unusually higher within the mixture; HBA basicity ($\beta$) never showed this so-called 'hyper-effect' or 'synergism'.[17,27] It is clear that the mixing of Reline with TEG is rendering one or more of the $NH_2$/C=O of urea, OH/$Cl^-$ of choline chloride, and OH/$OCH_2$ of TEG better H-bond accepting basicity especially closer to the equimolar region.

For a given composition of the (Reline + TEG) mixture, $E_T^N$ is found to decrease with increasing temperature in the

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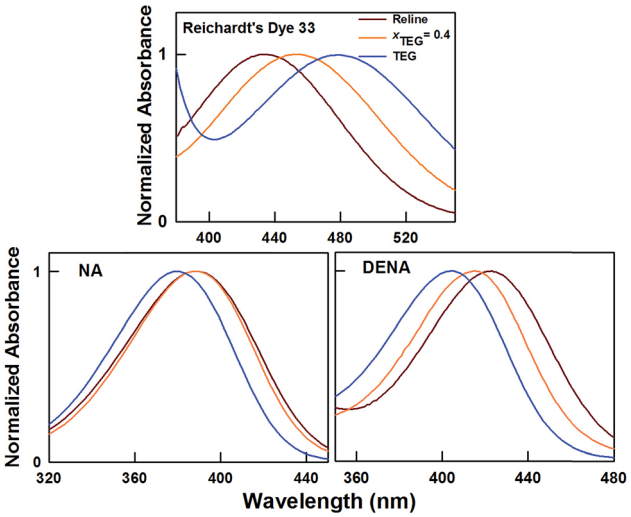



Fig. 1 Representative normalized absorbance spectra of Reichardt's dye 33 (250 μM), nitroaniline (NA, 10 μM), and *N*,*N*-diethyl-4-nitroaniline (DENA, 25 μM) in Reline, TEG, and (Reline + TEG) mixture having 0.4 mole fraction TEG at 298 K.

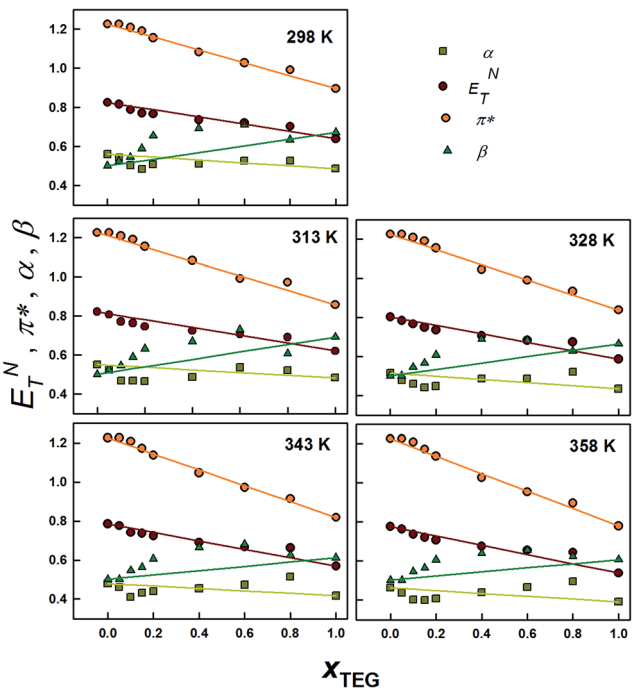



Fig. 2 Variation of $E_T^N$, $\pi^*$, $\alpha$, and $\beta$ with TEG mole fraction ($x_{TEG}$) within (Reline + TEG) mixtures as the temperature increases from 298 K to 358 K. Solid lines represent ideal-additive behavior. Error in all four parameters is ≤±0.01.

temperature range 298 to 358 K (Table S1†). A decrease in $E_T^N$ implies a decrease in the dipolarity/polarizability and/or the HBD acidity of the medium. In general, 'polarity' is suggested to usually decrease with increasing temperature due in major part to the increased average thermal reorientation of the dipoles.[13,30,31] This results in a decrease in dielectric constant with increasing temperature of polar liquids due partly to the destruction of the cooperative effect. For example, static dielectric constant of water is observed to decrease as the temperature is increased to 373 K.[32,33] It is interesting to note that the decrease in $E_T^N$ is twice in neat TEG as compared to that in neat Reline in the investigated temperature range. This is a direct consequence of the fact that while the decrease in $\alpha$ with increasing temperature is similar for Reline, TEG, and their mixture, $\pi^*$ of neat Reline does not change with temperature at all; it decreases significantly for neat TEG. Subsequently, the mixture shows decrease in $\pi^*$ with increasing temperature that is less than that observed for neat TEG. Similarly, $\beta$ for neat Reline also does not change with increasing temperature whereas for neat TEG it does decrease. The fact that Reline resists decrease in dipolarity/polarizability as well as HBA basicity as temperature of the medium is increased is manifested to some extent in the (Reline + TEG) mixtures where the decrease in these parameters is to a lesser extent as compared to TEG or other aqueous/nonaqueous solvents.

### 3.2. Fluorescence polarity probe behavior

**3.2.1 Dipolarity probes pyrene and pyrene-1-carboxaldehyde.** The pyrene solvent polarity scale (Py $I_1/I_3$) is defined by the ratio of its emission intensities, $I_1/I_3$, where $I_1$ is the intensity of the solvent-sensitive band corresponds to $S_1(v = 0) \rightarrow S_0(v = 0)$ transition and $I_3$ corresponds to the solvent-insensitive $S_1(v = 0) \rightarrow S_0(v = 1)$ transition.[34–36] The $I_1/I_3$ ratio increases with increasing solvent dipolarity and is a function of both the solvent dielectric ($\varepsilon$) and the refractive index ($n$) *via* the dielectric cross term, $f(\varepsilon,n^2)$.[35]

Experimentally obtained Py $I_1/I_3$ along with the ideal additive values (shown as dashed curves) estimated using eqn (6) proposed by Acree and coworkers[37] are presented in Fig. 3A.

$$\frac{I_1}{I_{3,\text{calc}}} = \frac{(I_{1,\text{Reline}} \times x_{\text{Reline}}) + (I_{1,\text{TEG}} \times x_{\text{TEG}})}{(I_{3,\text{Reline}} \times x_{\text{Reline}}) + (I_{3,\text{TEG}} \times x_{\text{TEG}})} \quad (6)$$

Irrespective of the temperature, Py $I_1/I_3$ in neat Reline is expectedly higher than that in TEG as Reline is constituted of ionic salt choline chloride involved in extensive H-bonding interactions with HBD urea. As TEG is added to Reline, initially the Py $I_1/I_3$ decreases sharply (in the Reline-rich region); the decrease becomes less sharp for $x_{TEG} \geq 0.2$. It is interesting to note that for $x_{TEG} \geq 0.2$, the Py $I_1/I_3$ are closer to their ideal additive values, however, at Reline-rich compositions, the Py $I_1/I_3$ are lower than those predicted from ideal additive behavior (expect for the highest temperature of 358 K). Addition of small amount of TEG lowers the dipolarity of the medium to a significant extent. The trend in the change in Py $I_1/I_3$, in this regard, is similar to those observed for $E_T^N$ and $\alpha$ (Fig. 2). These observations could be tentatively ascribed to the possible changes in the medium due to the solvent–solvent interactions leading to altered H-bonding capability; preferential solvation of the probe by TEG may also contribute to this observation. As expected, the dipolarity as reflected by the Py $I_1/I_3$ of the (Reline + TEG) mixture decreases as the temperature of the system is increased from 298 K to 358 K.


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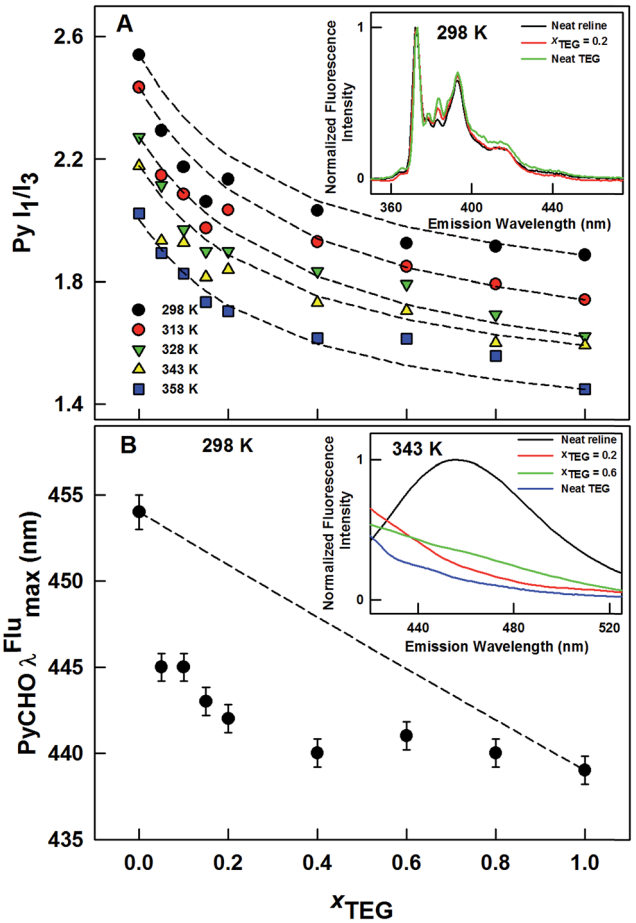



Fig. 3 Variation in pyrene [Py, 1 μM, $\lambda_{excitation}$: 337 nm, slit width (excitation/emission): 1/1 nm] $I_1/I_3$ as the temperature is increased from 298 K to 358 K (panel A) and pyrene-1-carboxaldehyde [PyCHO, 10 μM, $\lambda_{excitation}$: 365 nm, slit width (excitation/emission): 6/6 nm] $\lambda^{Flu}_{max}$ (panel B) at 298 K with TEG mole fraction ($x_{TEG}$) within (Reline + TEG) mixtures. Error in Py $I_1/I_3$ is ≤±0.02. Dashed curves represent ideal-additive behavior (insets of panels A and B show normalized fluorescence emission spectra of pyrene at 298 K and that of PyCHO at 343 K).

In fluid medium, fluorescence from pyrene-1-carboxaldehyde (PyCHO) can originate from either or both of the two closely lying excited singlet states, n–π* and π–π*. The π–π* is more stabilized and is brought below the n–π* on increasing the polarity of the surrounding thus rending π–π* the emitting state in more polar media. The emission from π–π* is manifested by a broad, reasonably intense emission that red shifts with increasing solvent dielectric.[38]

We found the fluorescence band representing emission from π–π* state to be very sensitive to the temperature for (Reline + TEG) system. While the lowest energy fluorescence emission band is readily observed at 298 K, at higher temperatures as the medium dipolarity is decreased instead of a band a shoulder appears rendering estimation of lowest energy fluorescence band maxima difficult (inset of Fig. 3B). Nonetheless, experimentally obtained lowest energy fluorescence emission maxima ($\lambda^{Flu}_{max}$) of PyCHO at 298 K along with the ideal additive values obtained using

$$(\lambda^{Flu}_{max,calc})^{-1} = [(\lambda^{Flu}_{max,Reline})^{-1} \times x_{Reline}] + [(\lambda^{Flu}_{max,TEG})^{-1} \times x_{TEG}] \quad (7)$$

are presented in Fig. 3B.

The response of PyCHO within (Reline + TEG) mixture at 298 K nicely corroborates that of pyrene and Reichardt's dye. As expected, the probe cybotactic region experiences significantly higher dipolarity in neat Reline as compared to that in neat TEG; PyCHO $\lambda^{Flu}_{max}$ decreases 15 nm. However, as small amount of the cosolvent TEG ($x_{TEG}$ = 0.05) is added to Reline the dipolarity as indicated by PyCHO decreases dramatically (decrease is 9 nm). Further addition of TEG results in very gradual decrease in the dipolarity of the medium. These outcomes are indicative of the fact that the probes PyCHO, due to its nonpolar character, may be preferentially solvated by TEG: the more nonpolar of the constituents in the mixture. However, as hypothesized earlier, changes in the H-bonding capabilities of the mixture due to the mixing of Reline and TEG resulting in unusually higher HBD basicity may be the key contributor to this. The overall dipolarity around PyCHO cybotactic region decreases markedly as a result of TEG over-crowding the probe and decreased polarity of the medium due to solvent–solvent interactions altering H-bond within the mixture. This lowering in the dipolarity combined with the lowering of the polarity of the medium due to temperature increase results in disappearance of the PyCHO fluorescence band representing emission from π–π* state.

**3.2.2 Ionic photoinduced charge-transfer probes TNS and ANS.** The major reason for the bathochromic shift in fluorescence emission band of probes TNS and ANS with increase in the polarity of the cybotactic region is the change in the intramolecular charge-transfer process within the probe.[39–41] However, along with solvent polarity, specific solute–solvent interactions, change in molecular conformation, intersystem crossing to the triplet state and mono-photonic photoionization might also take part in changing the fluorescence behavior of these probes.[39–41]

The $\lambda^{Flu}_{max}$ of TNS and ANS, respectively, obtained experimentally in (Reline + TEG) mixture along with the ideal additive values estimated using eqn (7) at different temperatures in the range 298–358 K are presented in Fig. 4. As expected, both TNS and ANS cybotactic regions are also more polar in Reline as compared to that in TEG. However, TNS appears to be more sensitive to the changes in the dipolarity of the (Reline + TEG) mixture as, irrespective of the medium temperature, the slope of $\lambda^{Flu}_{max}$ *versus* $x_{TEG}$ for TNS is higher than that of ANS. It is convenient to learn that as TEG is added to Reline in the Reline-rich region, responses of both TNS and ANS are suggestive of the negative deviation from ideal additive behavior as also manifested through the responses of Reichardt's dye, pyrene, and PyCHO discussed earlier. At Reline-rich compositions, either TNS and ANS are preferentially solvated by TEG or the changes in H-bonding due to mixture formation thus reducing the mixture polarity further than that expected or a combination of the two is again proposed to be the reason for these observations. We suspect the latter to be the dominant cause as preferential solvation of ionic probes by the nonpolar

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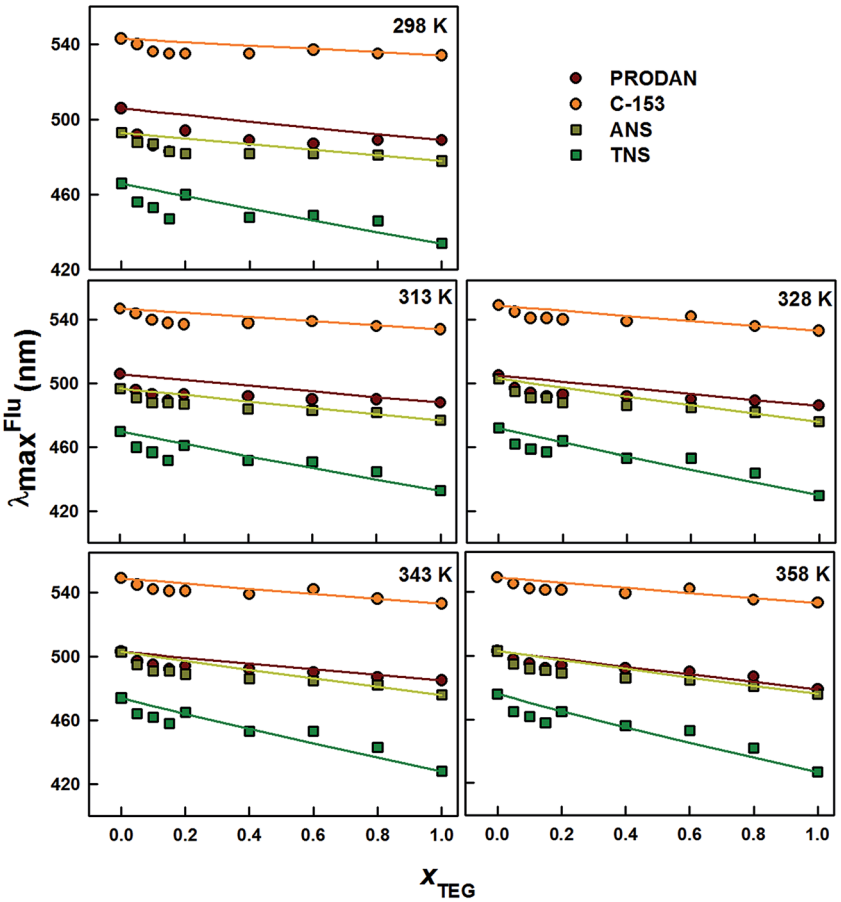



Fig. 4 Variation in $\lambda_{max}^{Flu}$ of 1-anilino-8-naphthalenesulfonate (ANS, 10 μM, $\lambda_{excitation}$: 346 nm), *p*-toluidinyl-6-naphthalene sulfonate (TNS, 10 μM, $\lambda_{excitation}$: 320 nm), 6-propionyl-2-(dimethylaminonaphthalene) (PRODAN, 10 μM, $\lambda_{excitation}$: 351 nm) and coumarin-153 (C-153, 10 μM, $\lambda_{excitation}$: 394 nm) with TEG mole fraction ($x_{TEG}$) for (Reline + TEG) mixtures at different temperatures ranging from 298 K to 358 K. Solid lines represent ideal-additive values.

constituent TEG as opposed to the polar-ionic constituent Reline is difficult to comprehend.

**3.2.3 Neutral photoinduced charge-transfer probes PRODAN and coumarin-153.** PRODAN undergoes substantial change in its dipole moment upon excitation with approximately no conformational change owing to intramolecular charge-transfer from an electron-donating dimethylamino group to electron-withdrawing propionyl group connected through an aromatic spacer.[41] Since PRODAN has no permanent charge it readily avoids contributions from ionic interactions.[41–43] PRODAN $\lambda_{max}^{Flu}$, as expected, is observed to be higher in neat Reline than in neat TEG at all investigated temperatures (Fig. 4). As TEG is added to Reline, the decrease in PRODAN $\lambda_{max}^{Flu}$ is more than that expected from the ideal additive behavior. This is in accord with that observed for Reichardt's dye, pyrene, PyCHO, ANS, and TNS. Further, the negative deviation from the ideal additive value is more in the Reline-rich region. It is also noteworthy that as the temperature of the system is increased, the deviation from the ideal additive value decreases. This temperature dependent behavior better supports the presence of Reline–TEG interactions leading to changed H-bonding as a result of mixing the two components as opposed to the preferential solvation of the probes by TEG as increase in temperature, in general, reduces H-bonding interactions within fluidic media.

The response of another similar neutral probe coumarin-153 (C-153) also corroborates the observations from Reichardt's dye, pyrene, PyCHO, ANS, TNS, and PRODAN. The $\lambda_{max}^{Flu}$ of C-153 within neat TEG is also lower than that observed within neat Reline, and the $\lambda_{max}^{Flu}$ are lower than that expected from ideal additive behavior for initial addition of TEG to Reline ($x_{TEG}$ < 0.4) (Fig. 4). The C-153 behavior further supports the proposition of preferential solvation of the probe by TEG and/or the presence of Reline–TEG interactions resulting in altered dipolarity of the mixture.

### 3.3. Density and dynamic viscosity measurements

Composition dependence of the bulk properties, density and dynamic viscosity, of the (Reline + TEG) mixture can reveal information regarding solvent–solvent interactions present within the mixture. This approach is considered non-invasive in nature as no external probe is added to obtain the insight into the interactions, if any, present between Reline and TEG.

Density of Reline, TEG, and their mixtures are measured in the temperature range 298–358 K in 15 deg interval (Table S2†).

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Irrespective of the temperature, density of neat Reline is found to be higher than that of neat TEG (our densities are in good agreement with those reported earlier for neat Reline and neat TEG[44]), and the density of Reline decreases monotonically as TEG is added. In order to afford the extent of interactions within (Reline + TEG) mixtures, we estimated excess molar volume ($V^E$) from experimental density data using the relationship

$$V^E = \frac{(x_{Reline} M_{Reline} + x_{TEG} M_{TEG})}{\rho_m} - \left(\frac{x_{Reline} M_{Reline}}{\rho_{Reline}} + \frac{x_{TEG} M_{TEG}}{\rho_{TEG}}\right) \quad (8)$$

here, $\rho_m$ is the density of the mixture and $M_{Reline}$ and $M_{TEG}$ are the molecular weights of Reline and TEG, respectively. The molecular weight of Reline was calculated from their individual components according to the equation:[6]

$$M_{Reline} = x_{choline\ chloride} M_{choline\ chloride} + x_{urea} M_{urea} \quad (9)$$

Molar volumes of neat TEG and neat Reline are 173.33 $cm^3\ mol^{-1}$ and 72.48 $cm^3\ mol^{-1}$, respectively, at 298 K. The $V^E$ at each temperature for (Reline + TEG) mixtures are presented as a function of $x_{TEG}$ in Fig. 5A. It is clear that $V^E$ are negative and are significant at each temperature throughout the entire composition range for all (Reline + TEG) mixtures (the absolute $V^E$ for Reline mixtures with TEG are significantly more than those with water[10]). Interestingly, as the TEG is added to Reline, the $V^E$ decreases sharply and reaches its minima value at $x_{TEG} \sim 0.2$; the maximum absolute $V^E$ for (Reline + TEG) mixture is observed in the range $0.2 \leq x_{TEG} \leq 0.6$. The temperature dependence of $V^E$, however, is not very significant. Probe responses from (Reline + TEG) mixtures discussed in the last two sections are also observed to be not too dependent on the temperature. The $V^E$ were fitted to the Redlich–Kister type polynomial expressions.[45] According to combined nearly ideal binary solvent/Redlich–Kister (CNIBS/R–K) model, the $V^E$ in a binary solvent mixture at a constant temperature can be expressed as:

$$V^E = x_{Reline} x_{TEG} \sum_{j=0}^{k} A_j (x_{Reline} - x_{TEG})^j \quad (10)$$

where $A$ and $k$ are the equation coefficients and the degree of the polynomial expansion, respectively. The numerical values of $k$ can be varied to find an accurate mathematical representation of the experimental data. Regression analysis was performed to fit the polynomials to our experimental density data and the results of the fit are shown in Fig. 5A as solid curves.

The negative $V^E$ generally points to contraction in volume upon mixing. The negative $V^E$ of (Reline + TEG) mixtures at all compositions hints at presence of relatively stronger inter-component H-bonding between Reline–TEG as compared to intra-component H-bonding between Reline–Reline and TEG–TEG. This would result in contraction in volume as the two components Reline and TEG are mixed. Facile interstitial accommodation of choline chloride and/or urea within H-bonded network of TEG or *vice versa* will also result in negative $V^E$. It is interesting to note, however, that the absolute value

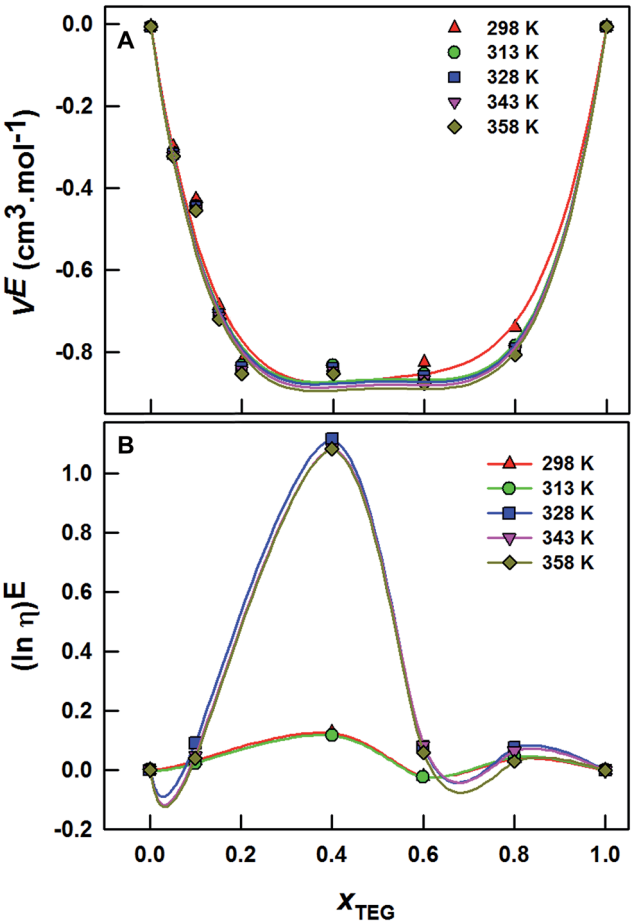



Fig. 5 Variations in excess molar volume ($V^E$ $cm^3\ mol^{-1}$, panel A) and excess logarithmic viscosities [$(\ln \eta)^E$, panel B] with TEG mole fraction ($x_{TEG}$) for (Reline + TEG) mixtures. Solid curves represent best fits according to the Redlich–Kister polynomial. Molar volumes of neat TEG and neat Reline are 173.33 $cm^3\ mol^{-1}$ and 72.48 $cm^3\ mol^{-1}$, respectively, at 298 K.

of $V^E$ does not change appreciably as the temperature is increased. As the system temperature is increased, we believe the reduction in inter-component H-bonding strength (which should result in decrease in absolute value of $V^E$) is partly offset by the increase in interstitial accommodation (which may result in increase in absolute value of $V^E$).

Dynamic viscosities ($\eta$) of (Reline + TEG) mixtures were experimentally measured over complete composition range at 15 K interval in the temperature range 298–358 K. As expected, monotonic decrease in dynamic viscosity is observed as the temperature is increased for a given composition of (Reline + TEG) mixture. In order to assess the interactions within (Reline + TEG) mixtures, excess logarithmic viscosities, $(\ln \eta)^E$, are estimated from the equation,[46]

$$(\ln \eta)^E = \ln \eta_m - [x_{Reline} \ln \eta_{Reline} + x_{TEG} \ln \eta_{TEG}] \quad (11)$$

where $\eta_m$ is the mixture viscosity, and are plotted as a function of $x_{TEG}$ in Fig. 5B (the solid curves represent the fit of the data according to Redlich–Kister polynomial expression). A careful examination of the plot reveals that, in general, $(\ln \eta)^E$ are

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positive irrespective of the temperature over the entire composition range. Further, $(\ln \eta)^E$ is found to be significant for $0.2 \leq x_{TEG} \leq 0.4$, which is in agreement with that observed for absolute value of $V^E$. Intensive H-bonding within the mixture between Reline and TEG may lead to positive $(\ln \eta)^E$. Based on the combined $V^E$ and $(\ln \eta)^E$ data it may be proposed that H-bonding interactions between Reline and TEG (*i.e.*, inter-species) are present within the mixture that are more prominent than the H-bonding between Reline (intra-species) and that between TEG molecules (intra-species). Contribution from the interstitial accommodation appears to be significant only at higher temperatures.[47–49] We envisage that this Reline–TEG H-bonding interactions manifest themselves through spectroscopic probe responses discussed earlier along with the unusual HBA basicity ($\beta$) of the mixture.

### 3.4. FTIR absorption and Raman spectroscopic observations

In order to further decipher the solute–solvent interactions from the solvent–solvent interactions present within (Reline + TEG) mixture and to pin-point the H-bonding functionalities/sites within the mixture, non-invasive (probe-free) approach of acquiring FTIR absorbance and Raman spectra of the mixture is undertaken next. As temperature does not appear to have important role in effecting properties of and interactions within (Reline + TEG) mixtures, FTIR absorbance and Raman spectra are acquired only at ambient conditions.

Fig. 6 presents FTIR absorbance spectra of C=O/$NH_2$ region (1300–1800 $cm^{-1}$, panel A) and of NH/OH region (3000–3800 $cm^{-1}$, panel B) for (Reline + TEG) mixtures of different compositions under ambient conditions (solid curves show experimental spectra whereas the dotted curves represent ideal-additive behavior). Our neat Reline FTIR absorbance bands are in good agreement with those reported in the literature.[50] A careful examination of the data reveals interesting features. The Reline band at 1607 and 1660 $cm^{-1}$ in C=O/$NH_2$ region and 3194 and 3326 $cm^{-1}$ in NH/OH region shifts considerably to higher energies as TEG is added to Reline – 1607 $cm^{-1}$ band shifts hypsochromically by 11 $cm^{-1}$ and 1660 $cm^{-1}$ band by 6 $cm^{-1}$, whereas 3194 $cm^{-1}$ band shifts hypsochromically by 13 $cm^{-1}$ and 3326 $cm^{-1}$ band by 16 $cm^{-1}$ upon addition of 0.6 mole fraction TEG. Interestingly, as indicated by the dotted spectra, these shifts are not predicted by the ideal-additive behavior (ideal-additive behavior predicts almost no shifts in these IR bands). For Reline, 1607 and 1660 $cm^{-1}$ bands are assigned to asymmetric and symmetric bending of $NH_2$ (*i.e.*, $\delta_{as}NH_2$ and $\delta_sNH_2$), respectively, and 3194 and 3326 $cm^{-1}$ to $NH_2$ symmetric bending/C=O stretching ($\delta_sNH_2/\nu C{=}O$) and $NH_2$ symmetric stretching ($\nu_sNH_2$), respectively. It is noted that the 3194 $cm^{-1}$ band is a combination band and not a mixed vibrational character. In support to the FTIR absorbance band shifts to the higher energy, Raman band at 3368 $cm^{-1}$ corresponding to $NH_2$ stretching also shifts hypsochromically by 27 $cm^{-1}$ to 3395 $cm^{-1}$ as 0.6 mole fraction of TEG is added to Reline (Fig. 7).

The hypsochromic shifts of different vibrational modes of $NH_2$ and/or C=O hint at weakening of the H-bonding involving these functionalities on Reline. These functionalities are present only on the urea part of the Reline. This observation gets further support from the probe behavior, especially the unusually high HBA basicity ($\beta$) observed for the (Reline + TEG)

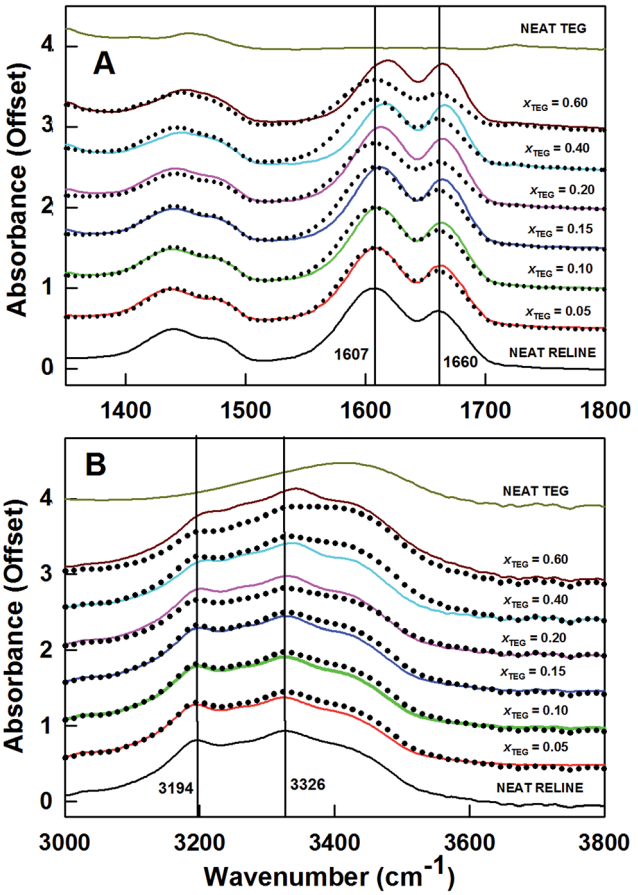



Fig. 6 FTIR absorbance spectra of (Reline + TEG) mixtures under ambient conditions: C=O/$NH_2$ region (panel A) and NH/OH region (panel B). Solid vertical lines illustrate the spectral shifts.

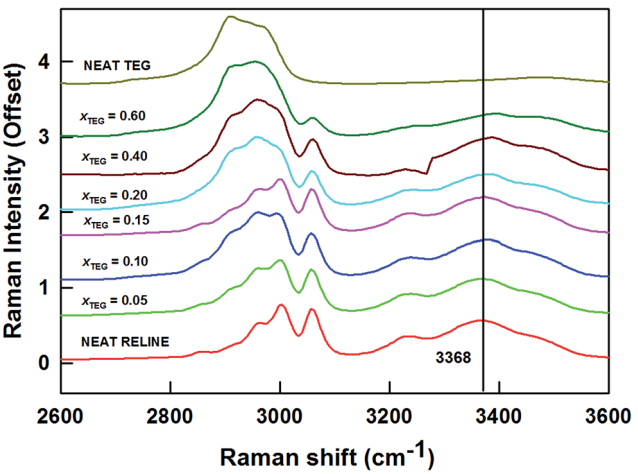
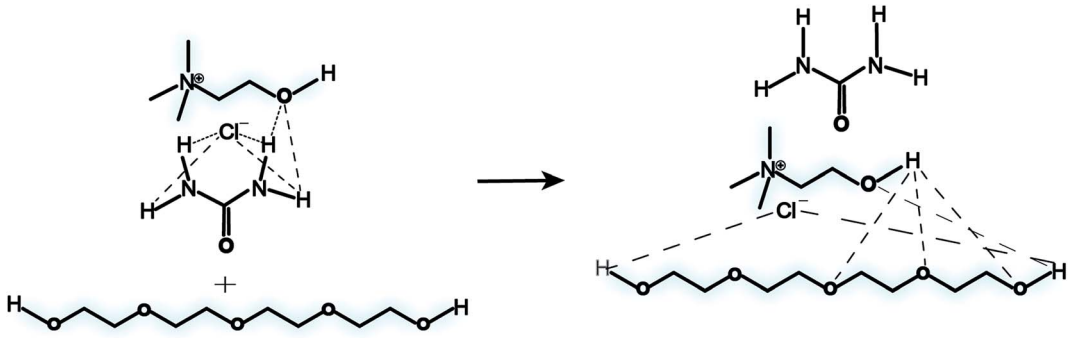



Fig. 7 Raman spectra ($\lambda_{excitation}$ = 532 nm) of (Reline + TEG) mixtures under ambient conditions. Solid vertical line illustrates the spectral shift.


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Scheme 2 Proposed representation of changes in H-bonding interactions upon formation of (Reline + TEG) mixtures.

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mixtures, which is probably due to lone pairs on nitrogen and/or oxygen of urea becoming more available to accept H-bonding. The chloride anion and the –OH functionality of choline chloride, on the other hand, appear to be involved in strong H-bonding interactions with hydroxyl termini and –O– of the ethoxy functionalities of TEG that results in negative excess molar volume and positive excess logarithmic viscosities (Scheme 2).

## 4. Conclusions

Although a prototypical DES Reline possesses interesting properties as it is formed by mixing two solids at room temperature, an ammonium salt choline chloride and an H-bond donor urea, its properties can be readily modified by adding an environmentally-benign cosolvent TEG which offers multiple H-bonding possibilities. Solvatochromic probe responses within TEG-added Reline highlight the complexity associated with the interactions present within this media. While dipolarity/polarizability and H-bond donating acidity do change as TEG is added to Reline, H-bond accepting basicity, $\beta$, for certain mixture compositions are found to be unusually high; even higher than its value in neat TEG which has higher $\beta$ of the two species forming the mixture. Fluorescence probe behavior further support the modification in the solvation properties as TEG is mixed with Reline. Excess molar volumes of (Reline + TEG) mixtures are negative irrespective of the mixture composition and temperature suggesting reduction in volume as a result of mixing the two species. This outcome combined with the positive excess logarithmic viscosities of the mixture leads us to propose the presence of stronger H-bonding interactions between Reline and TEG as compared to those between Reline and Reline and between TEG and TEG. FTIR absorbance and Raman spectroscopic data strongly indicate that within the mixture involvement of urea in H-bonding is decreased (this explains the unusually high H-bond accepting basicities of the mixture) leading us to believe that the H-bonding between choline chloride and TEG is increased substantially. Interesting H-bonding reversal due to mixing of TEG to Reline is highlighted by our results. These outcomes may open new avenues for DESs and their green cosolvent-modified nonaqueous mixtures in materials synthesis and analytical chemistry.

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